Research-to-operational hourly $PM_{2.5}$ forecasting at 27 Indonesian stations

Experimental model-development report — not yet an operational public product
04 September 2026

Scope. Daily 00 UTC forecasts at +3, +6, +12, +24, +48, and +72 hours. The output is predictive and does not provide causal or source attribution.

Contents

1 Executive finding

The validation-selected model is **CAMS model-output-statistics LightGBM**. Across the six lead times in the untouched 1 January–31 August 2026 test period, its mean station-balanced mean absolute error (MAE) is **9.82 μg m^{-3}**, corresponding to **+28.5%** mean skill relative to persistence and **+34.2%** relative to raw CAMS. Improvement occurs in **94.4%** of station–lead combinations, so aggregate skill does not imply uniform local benefit. The nominal 80% prediction intervals attain **78.8%** mean test coverage across leads.

Operational decision. Retain the system as pre-operational. It is suitable for scheduled shadow forecasts and scorecard monitoring, but public or duty-forecaster reliance should wait for prospective evaluation, explicit input-latency tests, a second forecast cycle, and an assessment of forecast meteorology.

2 Question and intended use

The question is whether a pooled machine-learning model can improve station-level hourly $PM_{2.5}$ forecasts over simple persistence at 27 BMKG locations while remaining computationally light enough for routine operation. One direct model is fitted for each forecast lead; direct forecasting avoids recursive error propagation. The intended output is a concentration forecast in μg m^{-3} plus a calibrated uncertainty interval at each observed station.

The primary inference domain is the existing network. A separate station-holdout experiment asks how well the approach transfers to a location absent from fitting. Neither experiment creates a continuous spatial concentration field.

3 Data, quality control, and provenance

The supplied archive contains **1,158,532** rows from 27 hourly station files spanning 2021-01-01 to 2026-08-31. After documented error screening, **1,116,030 (96.33%)** source rows contain valid $PM_{2.5}$. Exact duplicated station-hours were removed only in derived data; no conflicting duplicate key was accepted. Negative $PM_{2.5}$ and values at or above 985 μg m^{-3} were treated as instrument error values. Plausible extremes below the cutoff were retained.

Relative humidity outside (0,100]% and temperature outside -10 to 50 °C were made unavailable as predictors without altering $PM_{2.5}$. This screening identified a block of physically impossible temperature values at Koto Tabang; the values were not corrected or imputed. Median station-level valid $PM_{2.5}$ coverage within each station's observed span is **94.3%**.

CAMS global atmospheric-composition forecast $PM_{2.5}$ was extracted from the official `ECMWF/CAMS/NRT` Earth Engine mirror [7] and bilinearly sampled to station coordinates. Source mass density in kg m^{-3} was converted explicitly to μg m^{-3}. The parent CAMS service provides global forecasts twice daily on a 0.4° grid and is an atmospheric model product, not a surface observation [1]. The mirror is three-hourly, so +3 h is the earliest evaluated lead; no +1 h field was synthesized.

The extract retained **215,916 of 216,918 station–issue–lead records (99.54%)**. Missing records comprise a partial spatial gap on 17–23 February 2024 and a fully missing 00 UTC initialization on 30 June 2025; they were not imputed. A direct Copernicus archive check covered 20 in-domain stations and five common leads on 1 January 2026. Across 100 native grid-point comparisons, the maximum absolute difference was 0.000017 μg m^{-3}; station-resampling implementations differed by 0.107 μg m^{-3} on average and 1.217 μg m^{-3} at most. This supports identity of the underlying field while documenting interpolation-level differences. The bounded experiment uses CAMS $PM_{2.5}$ plus observed temperature and humidity histories; forecast meteorology is a declared next-stage input.

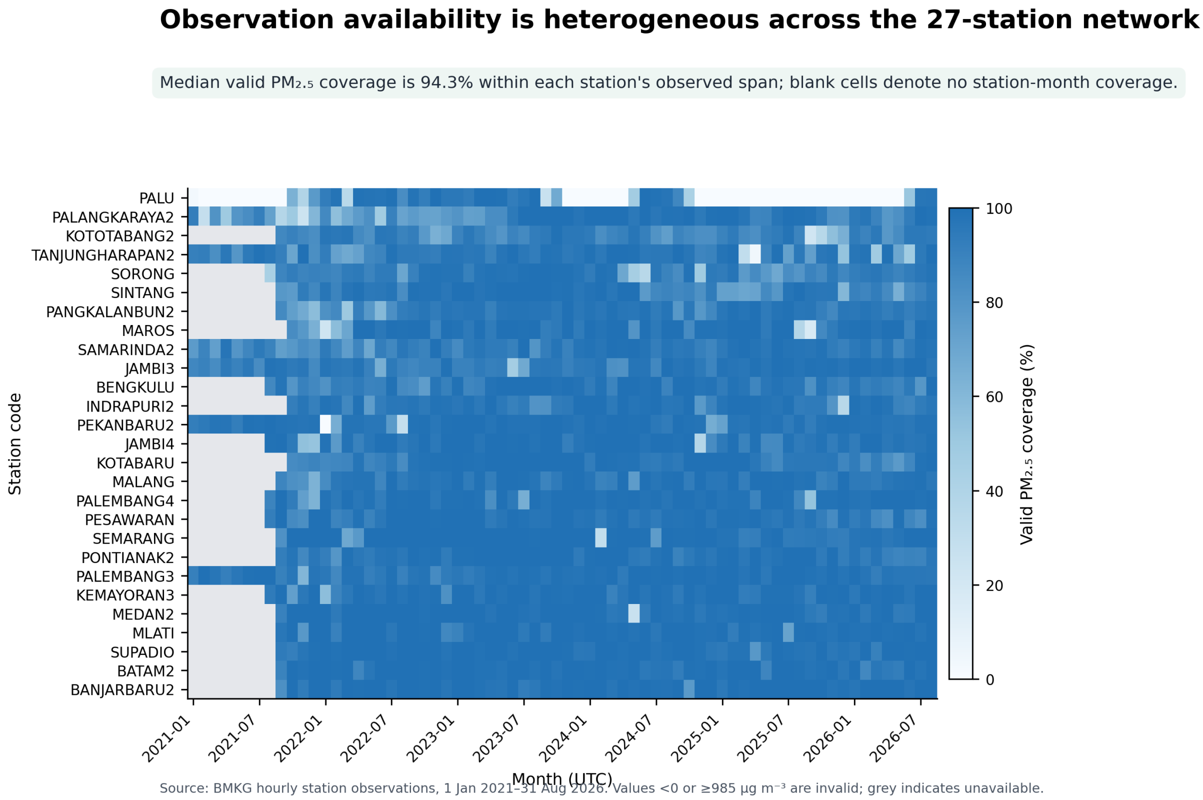

Figure 1: Monthly valid PM2.5 coverage across the 27 stations. Missing coverage is distinct from a valid zero concentration.

4 Prediction design

4.1 Inputs

Predictors use only timestamps at or before issue time: $PM_{2.5}$ lags from 0 to 168 h; temperature and relative-humidity lags to 24 h; rolling $PM_{2.5}$ mean, standard deviation, maximum, and availability over 3–168 h; local-hour and annual-cycle encodings; station coordinates and region; contemporaneous and distance-weighted network context within 400 km; and forecast-valid CAMS $PM_{2.5}$. Missing predictor values are retained for native tree handling. Feature importance is diagnostic and not causal importance.

4.2 Chronological separation

Targets from 2023–2024 form training, 2025 is used for point-model early stopping and model choice, and 2026 through 31 August is evaluated once as the final test. The uncertainty workflow adds a nested temporal separation: January–June 2025 tunes quantile tree counts, while July–December 2025 is reserved exclusively for conformal calibration. Splits are assigned by target time, not issue time. The derived table contains **216,918** station–issue–lead rows and **200,724** valid targets. Automated audits found 0 duplicate modeling keys, 0 non-future targets, and no target-time overlap among train, validation, and test. Chronological and station-blocked validation are used because random splitting of structured environmental data can underestimate predictive error [6].

4.3 Models and uncertainty

Persistence and a training-only station/local-time climatology are reference forecasts. Candidate learners are observation-only LightGBM, CAMS model-output-statistics (MOS) LightGBM, and CAMS MOS XGBoost. Gradient-boosted trees represent nonlinear interactions and missingness without the compute cost of deep sequence models [3, 4]. CAMS calibration with machine learning has prior scientific precedent, although performance is domain-specific [2].

Point and quantile predictions are clipped only at the physical lower bound of 0 µg m^{-3}. Quantile LightGBM estimates the 10th, 50th, and 90th conditional percentiles. Quantile crossing is removed by sorting, then a lead-specific split-conformal expansion is estimated from the held July–December 2025 calibration block pooled across stations. Neither fitting nor interval calibration accesses 2026. Conformalized quantile regression provides a principled basis for adaptive prediction intervals under exchangeability assumptions [5]; temporal autocorrelation and later CAMS system changes mean empirical monitoring remains necessary.

5 Validation selection and independent test results

Model choice used mean station-balanced validation MAE across all six leads. The specified tie-break prefers CAMS LightGBM when it lies within 1% of the minimum, reducing deployment complexity while retaining a physically based forecast input.

Candidate	Validation MAE	Skill vs persistence (%)	Selected
CAMS MOS LightGBM	7.12	28.7	Yes
CAMS MOS XGBoost	7.16	28.0	No
Observation-only LightGBM	7.19	28.3	No

The selected model's validation criterion is **7.12 µg m^{-3}**. The following table reports the subsequently opened test set. Positive bias means overprediction. The confidence interval is a 1,000-replicate station-week block bootstrap for absolute-error improvement over persistence.

Lead (h)	n	MAE	Persistence MAE	Raw CAMS MAE	Skill (%)	RMSE	Bias	r	95% CI: MAE gain
+3	5,918	7.62	12.64	9.95	36.5	16.48	-2.36	0.73	[3.79, 6.29]
+6	5,753	6.29	14.18	7.91	48.0	13.44	-1.58	0.72	[7.26, 9.59]
+12	5,897	10.50	16.69	17.61	32.9	16.62	-2.23	0.58	[5.77, 8.14]
+24	5,859	10.84	13.11	17.51	17.6	19.55	-3.43	0.66	[1.02, 2.45]
+48	5,839	11.66	13.90	18.10	16.9	21.39	-4.41	0.59	[0.99, 3.10]
+72	5,832	12.03	14.68	18.55	19.2	21.83	-4.45	0.57	[1.16, 3.50]

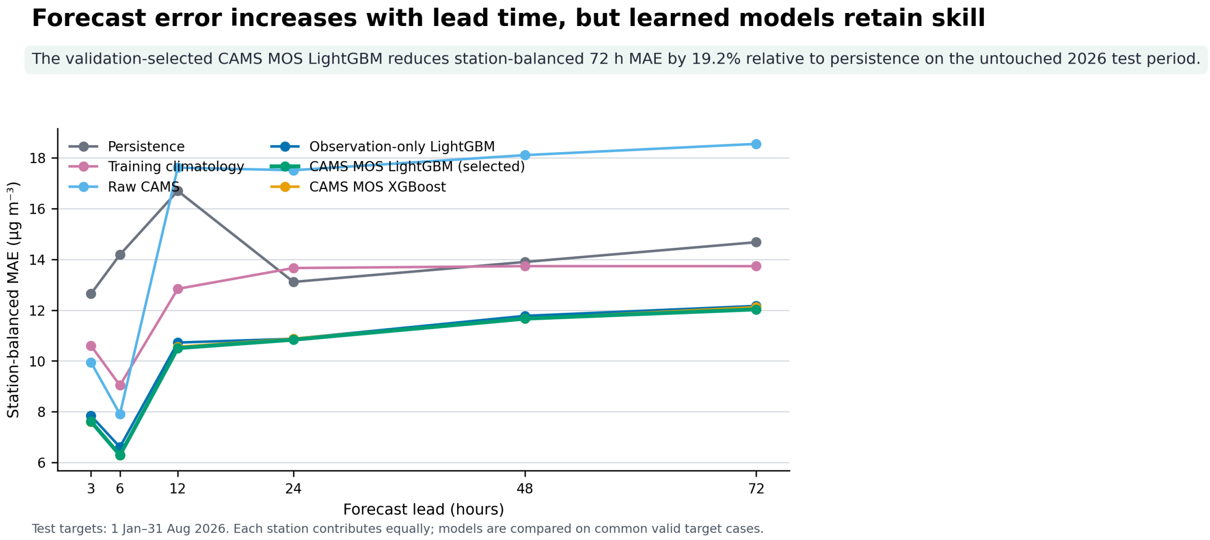

Figure 2: Station-balanced test MAE by lead for all candidates and reference forecasts.

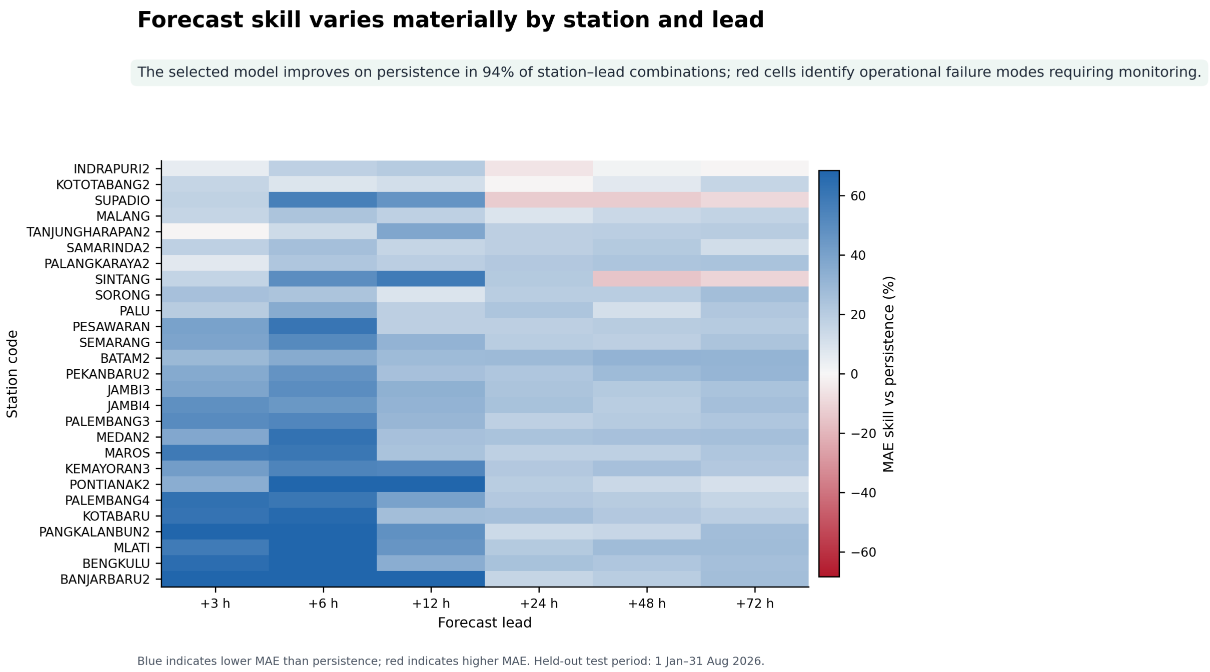

Figure 3: Selected-model MAE skill relative to persistence for every station and lead.

5.1 Chronological behaviour across development stages

Aggregate errors do not show whether the forecast follows episode timing, reacts late, or compresses peaks. Figure 4 therefore compares observed and predicted +24 h sequences in target-time order. Each point is the daily median across stations with valid data; no temporal smoothing or interpolation is applied.

Training-period values are not fitted predictions. They use expanding-window assessment blocks for July–December 2023, January–June 2024, and July–December 2024. Every assessment target is later than all targets used to fit its fold. The displayed model family and tree counts were selected later using 2025 validation, so this retrospective out-of-fold diagnostic describes temporal behaviour and is not an independent model-selection score. The 2025 panel is validation evidence and the 2026 panel remains the independent test.

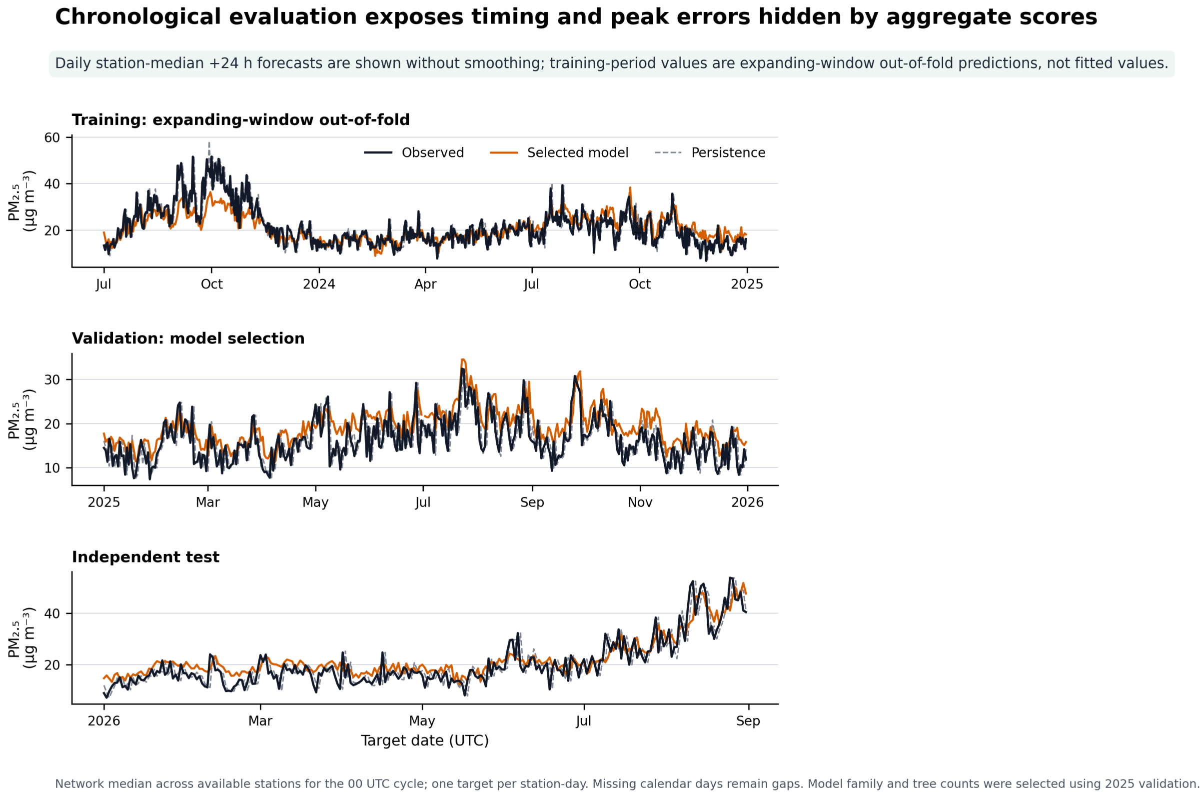

Figure 4: Chronological observed, selected-model, and persistence PM2.5 at +24 h for expanding-window out-of-fold training assessments, validation, and independent testing. Values are daily station medians for the 00 UTC forecast cycle.

The nine-page station atlas is integrated into Appendix B of this report. It shows complete independent-test sequences at +6, +24, and +72 h for all 27 stations, including calibrated 10th–90th percentile intervals, so poor-performing stations and short-lived episodes remain inspectable rather than concealed by national aggregation.

5.2 Incremental value of CAMS

Relative to the otherwise identical observation-only LightGBM, adding forecast-valid CAMS $PM_{2.5}$ reduced station-week-balanced MAE by **0.065 µg m^{-3}** in validation (95% bootstrap interval 0.034 to 0.098) and **0.178 µg m^{-3}** in the independent test (95% interval 0.119 to 0.242). This is an aggregate predictive association, not causal evidence. Test-period gains are clearest at +3 to +12 h; intervals cross zero at each of +24, +48, and +72 h, so longer-lead incremental benefit remains inconclusive individually.

Lead (h)	MAE gain	Relative gain (%)	95% CI	Station-weeks
+3	0.295	3.34	[0.194, 0.394]	945
+6	0.390	5.43	[0.286, 0.489]	944
+12	0.240	2.17	[0.165, 0.315]	944
+24	0.012	0.11	[-0.179, 0.184]	945
+48	0.045	0.36	[-0.057, 0.143]	945
+72	0.089	0.68	[-0.033, 0.222]	945
All leads	0.178	1.65	[0.119, 0.242]	945

The five least favourable station–lead combinations include SINTANG (-15.2%), SUPADIO (-13.1%), SUPADIO (-12.9%), SINTANG (-10.2%), SUPADIO (-9.5%). These are failure modes for investigation, not grounds for deleting observations or selectively omitting stations.

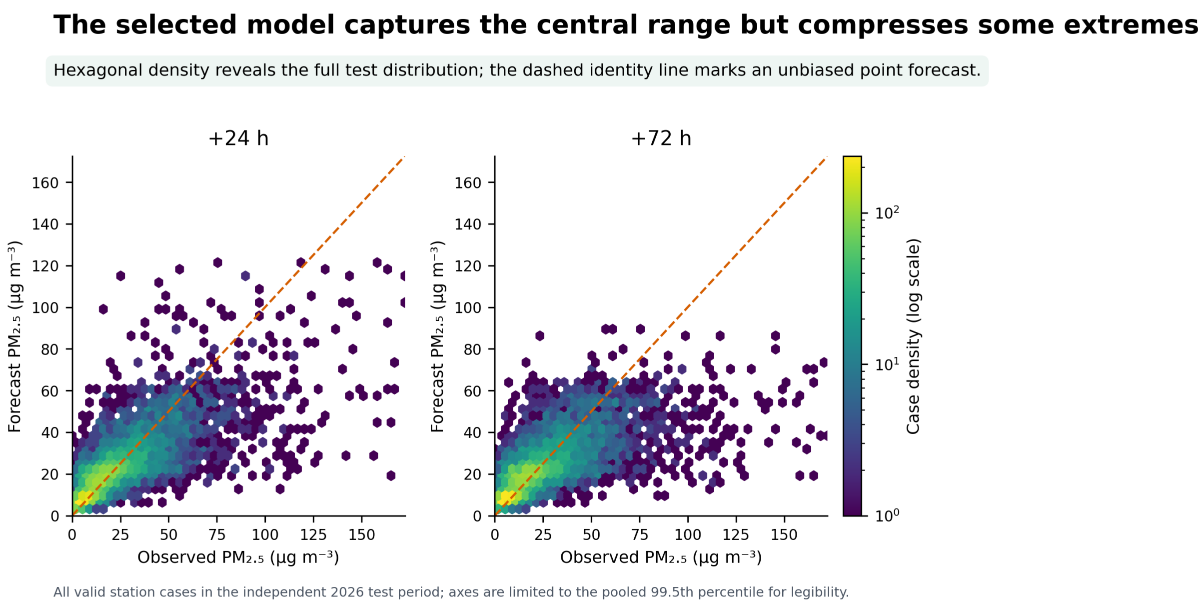

Figure 5: Observed versus selected-model PM2.5 at +24 h and +72 h. Display axes are limited to the pooled 99.5th percentile for legibility; metrics use the full valid range.

6 Uncertainty and high-concentration performance

Lead (h)	n	Coverage (%)	Mean width	Interval score
+3	6,050	79.4	19.8	42.3
+6	5,905	79.3	17.7	33.0
+12	6,108	81.1	34.0	53.0
+24	6,084	78.0	31.7	53.4
+48	6,084	77.5	32.3	57.6
+72	6,084	77.5	32.5	60.6

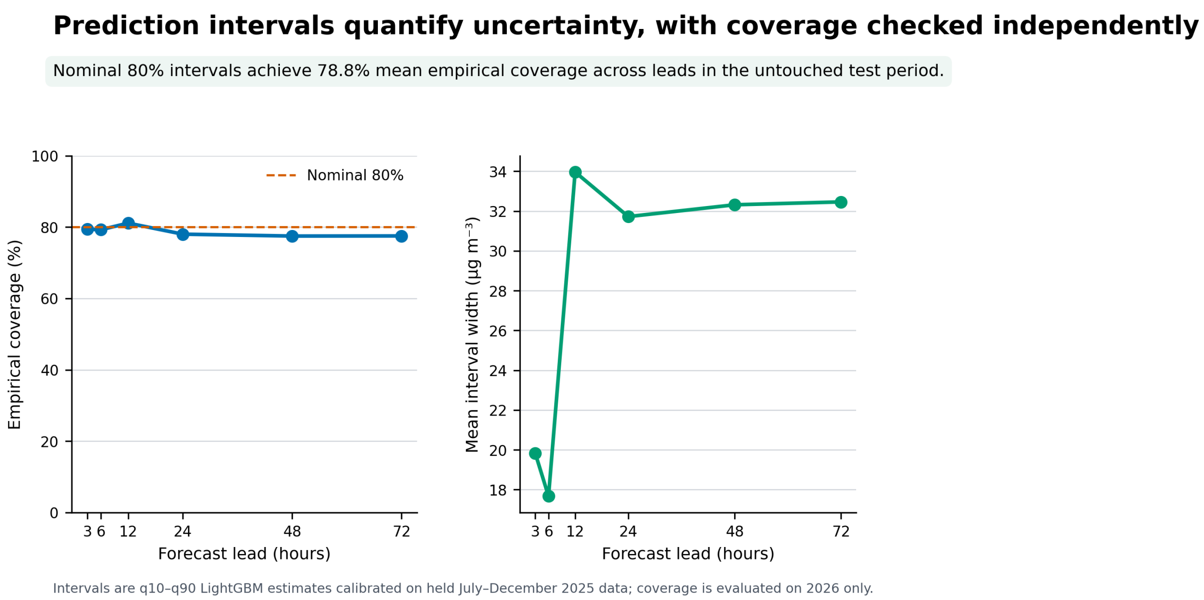

Figure 6: Independent empirical coverage and mean width of the nominal 80 percent intervals.

High-concentration events are defined separately for each station and lead using the station's training-period 90th percentile. Probability of detection (POD) is the fraction of observed events detected; false-alarm ratio (FAR) is the fraction of predicted events that did not occur; critical success index (CSI) penalizes misses and false alarms.

Lead (h)	Hits	Misses	False alarms	POD (%)	FAR (%)	CSI (%)
+3	392	322	148	54.9	27.4	45.5
+6	403	314	125	56.2	23.7	47.9
+12	225	475	85	32.1	27.4	28.7
+24	357	423	148	45.8	29.3	38.5
+48	280	500	153	35.9	35.3	30.0
+72	252	528	154	32.3	37.9	27.0

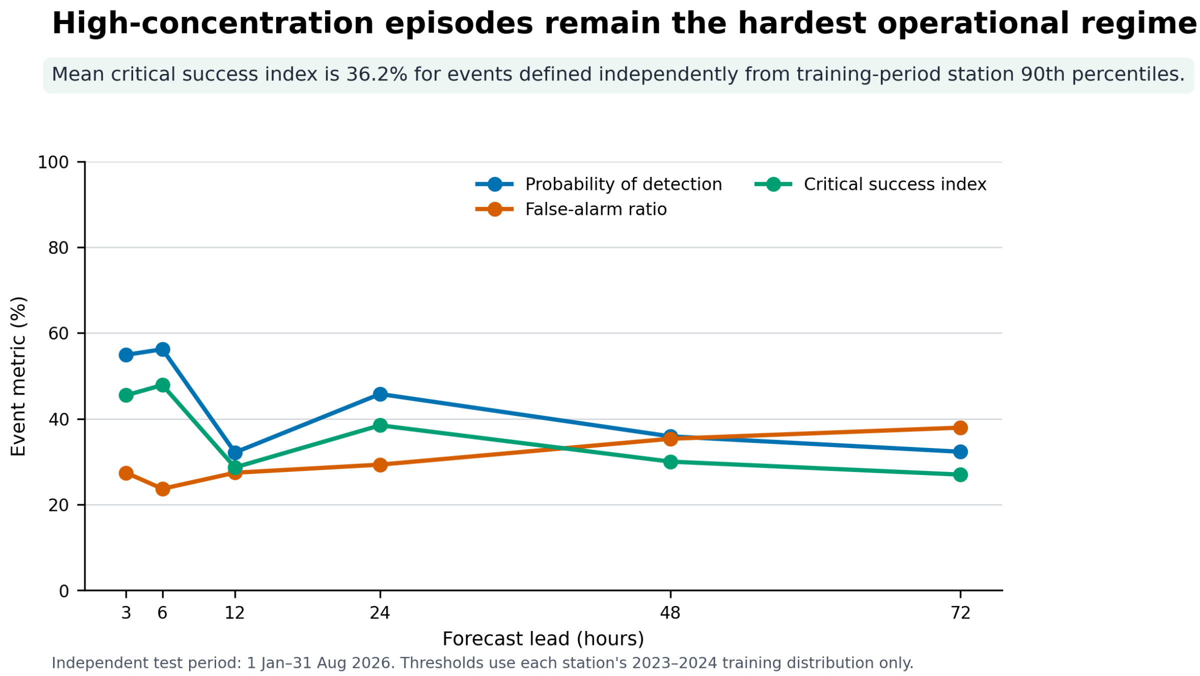

Figure 7: High-concentration event performance on the independent test period.

7 Robustness, spatial transfer, and interpretation

The same-network test estimates performance for stations represented in training. Five station folds provide a separate transfer diagnostic: each held station is tested with a model fitted to the remaining stations and without station identity. Fold assignment is independent of target values.

Lead (h)	Stations	MAE	Skill vs persistence (%)
+3	27	7.78	35.0
+6	27	6.50	46.4
+12	27	11.29	25.6
+24	27	11.09	14.1
+48	27	11.98	13.0
+72	27	12.42	14.4

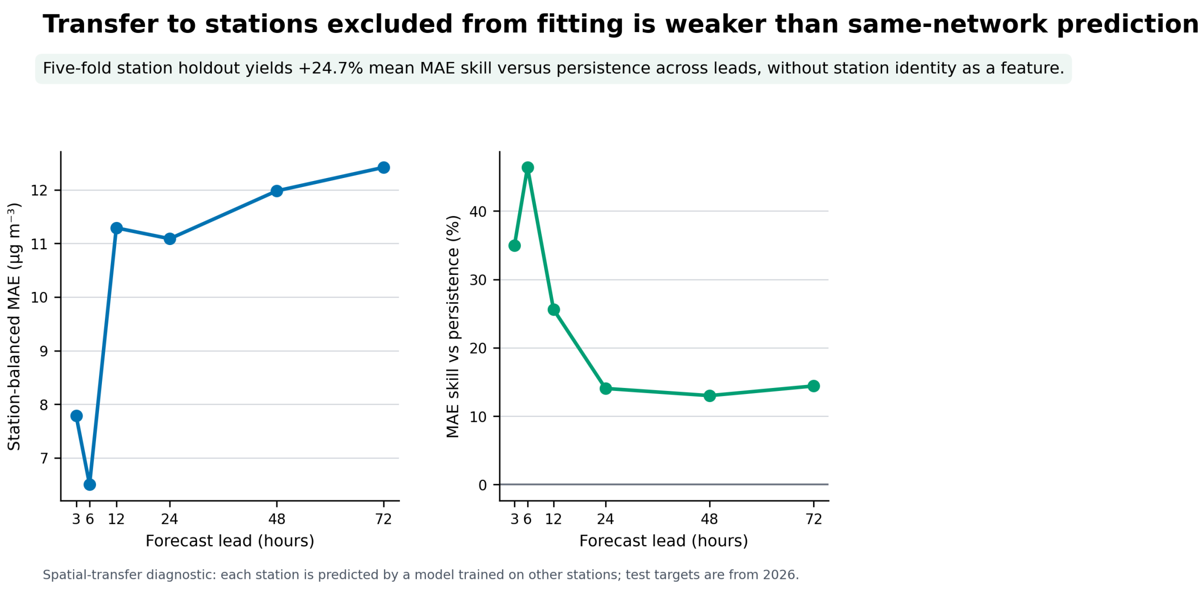

Figure 8: Performance when test stations are excluded from model fitting.

The 2024-only training sensitivity holds model form and tree counts fixed, changing only the fitting window. A positive difference means the shorter window has higher test MAE.

Lead (h)	Frozen MAE	2024-only MAE	Difference
+3	7.62	7.85	0.24
+6	6.29	6.63	0.34
+12	10.50	10.99	0.49
+24	10.84	11.28	0.44
+48	11.66	12.01	0.35
+72	12.03	12.39	0.36

Distribution-shift diagnostics compare predictor missingness, standardized mean differences, and population stability index between training and test. These statistics identify monitoring candidates; they do not establish that a shift caused an error. Residual bias is stratified by station, target month, and local target hour.

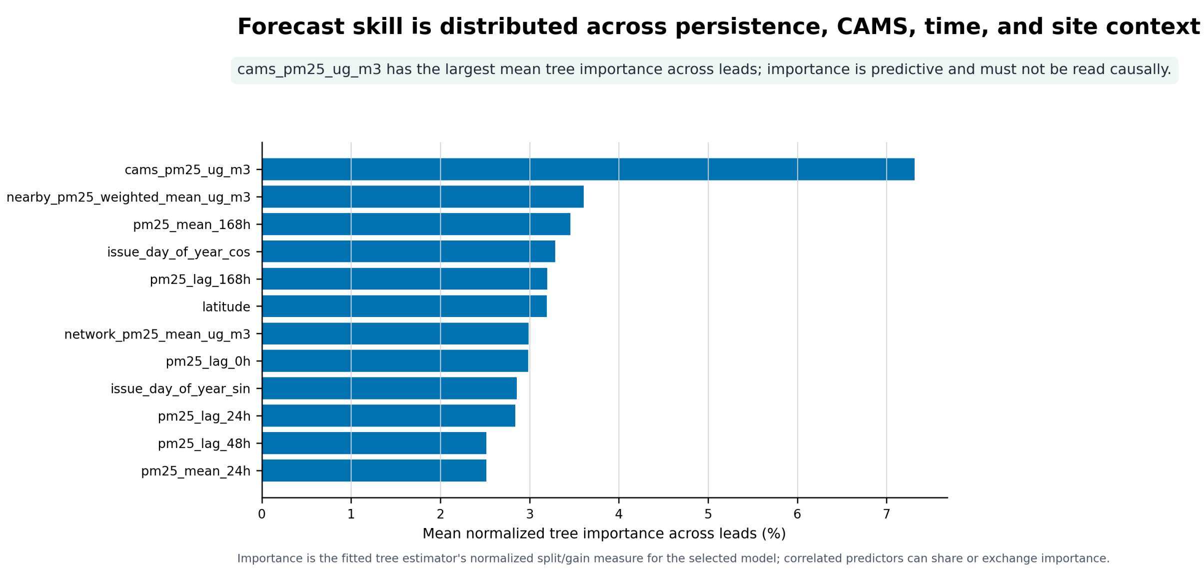

Figure 9: Mean fitted-tree feature importance for the selected model. Correlated predictors can exchange importance.

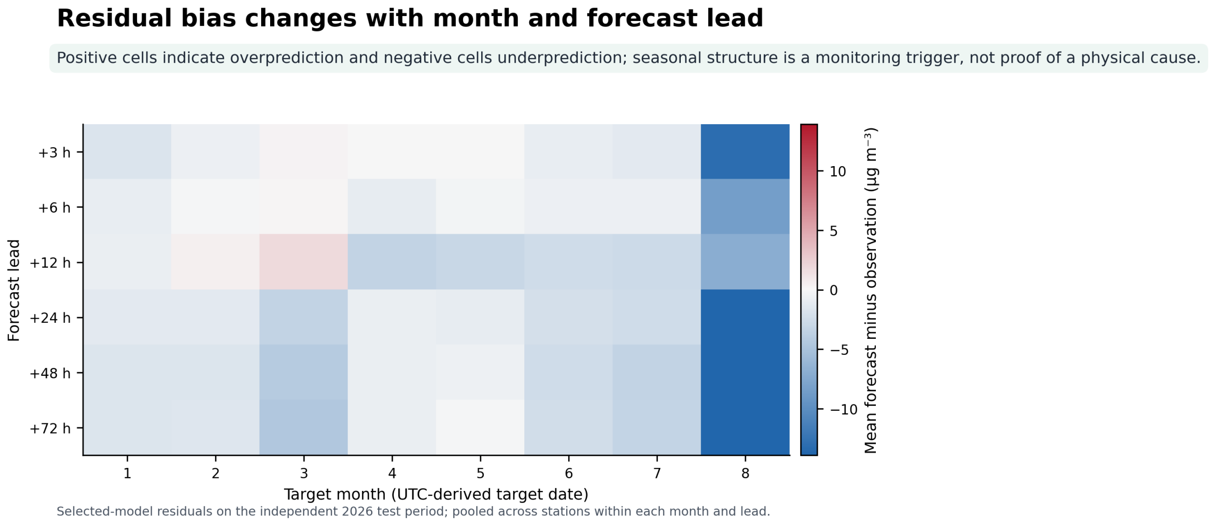

Figure 10: Selected-model mean residual by target month and lead.

8 Computational requirements and operational design

The experiment is designed for a CPU workstation; no GPU is required. Training uses at most 8 CPU threads. Measured peak memory was **preparation 1.43 GiB, training 0.81 GiB**. Measured preparation time was **92.9 s**, candidate/uncertainty/transfer evaluation was **103.5 s**, and deployment refitting was **24.4 s**. The complete deterministic pipeline took **4.7 min** across its latest successful stage measurements. The serialized deployment bundle is **9.8 MiB**. Warm inference across all point, fallback, and interval models totals about **0.136 s**, excluding model loading, input download, and feature preparation. The measured prepared-input end-to-end command took **2.13 s** for 162 station–lead rows.

For a routine 00 UTC run, a practical allocation is **4–8 CPU cores, 4 GiB RAM, and 2 GiB working storage**. Model inference should finish in under one minute; CAMS acquisition is network- and service-limited and should be budgeted at **5–30 minutes** with retries. The initial 3.7-year Earth Engine backfill required **12.7 min** of measured server-query time. A full deterministic research rebuild should be budgeted at **5–15 minutes** on a comparable CPU workstation; slower CPUs and uncached report dependencies may require longer.

Deployment point models are refitted through December 2025; deployment quantile models stop at June 2025 so July–December remains a held calibration block. If CAMS $PM_{2.5}$ is missing, a separately fitted observation-only point forecast is used and the status is marked degraded; calibrated intervals are withheld. Observation older than six hours is explicitly flagged. This fallback supports continuity but must not be presented as equivalent quality.

8.1 Implemented daily shadow workflow

The non-public shadow workflow is scheduled daily at 17:15 WIB (10:15 UTC). It saves an immutable BMKG dashboard snapshot, freezes the observation cutoff, acquires the current CAMS 00 UTC initialization directly from the Copernicus archive, writes atomic forecasts with hashes and freshness/status fields, and scores them only after observations appear. First-seen station-hour values are retained for verification and later raw snapshots preserve revisions. There is no public upload.

The first end-to-end engineering run on 4 September 2026 generated **162 rows in 67.7 s** with **0 degraded rows**. It produced **108 prospectively eligible rows** and labelled **54 rows** whose target times had already occurred. This first execution is a workflow test, not a prospective performance result.

That engineering run completed **11.3 h after the 00 UTC initialization**. Observation freshness in shadow output is evaluated relative to actual generation time as well as model initialization; otherwise an observation timestamped at 00 UTC would be incorrectly described as fresh about eleven hours later.

1. Retrieve and validate the 00 UTC CAMS forecast after publication.
2. Freeze the latest observation cutoff and record station freshness.
3. Construct predictors without accessing later observations.
4. Produce forecasts, intervals, status flags, hashes, and logs.
5. Score forecasts when observations arrive; retain missing cases rather than silently backfilling.
6. Record warnings for missing CAMS, stale station data, schema/version changes, degraded operation, extreme residuals, and interval undercoverage.

9 Limitations and readiness gates

- One daily cycle was retrospectively tested; this is not evidence for 12 UTC or arbitrary issue times.
- Direct CAMS availability is later than model initialization. At the installed schedule, +3 h and +6 h are latency diagnostics rather than prospective forecasts; +12 h is eligible only if generation completes before 12 UTC. Prospective reporting uses the stored per-row eligibility flag.
- The test covers eight months of 2026, not a complete annual cycle, and no prospective shadow period has yet been observed.
- CAMS is much coarser than a station and its forecast system can change; a sampled grid value is not a station measurement.
- The CAMS mirror is 99.54% complete for the specified station–issue–lead grid; missing cycles are excluded from primary complete-case comparisons and require the declared degraded fallback in operations.
- Forecast meteorology was not included. Observed meteorology at or before issue time cannot represent future dispersion conditions.
- Missingness, maintenance, relocation, calibration change, and reporting latency can alter station performance.
- Lead-specific conformal calibration pooled across stations assumes exchangeability more strongly than this autocorrelated, spatially dependent archive warrants. The later calibration block prevents fitting leakage, but held-out coverage is reported rather than guaranteed operationally.
- Station-transfer results do not validate a continuous spatial map or arbitrary new station.
- Feature importance is not emissions, wildfire, transport, contribution, or causal attribution.

Before public operations, require at least 60–90 days of prospective shadow scoring; a documented data-arrival cutoff; 12 UTC backtesting if desired; incremental-skill tests for forecast wind, boundary-layer height, humidity, and precipitation; CAMS-version monitoring; station-specific fallback rules; alerting and rollback; and scientific sign-off on thresholds and wording.

10 Reproducibility and data-quality detail

Every source observation file, external archive, sampled field, configuration, research model, and deployment model has a SHA-256 checksum in the provenance record. Random seeds are fixed. Derived tables preserve UTC timestamps, source station identity, explicit units, target lead, split, and missingness. No source observation was changed.

Station	Name	Start	End	Valid coverage (%)	Absent hours	Invalid T
PALU	Lore Lindu Bariri	2021-01-23	2026-08-31	42.6	25,203	0
PALANGKARAYA2	Palangka Raya	2021-01-01	2026-08-31	84.6	4,564	0
KOTOTABANG2	Koto Tabang	2021-09-24	2026-08-31	85.7	1,805	3,662
TANJUNGHARAPAN2	Tanjung Harapan	2021-01-01	2026-08-31	88.1	4,513	0
SORONG	Sorong	2021-08-19	2026-08-31	88.9	2,250	0
SINTANG	Sintang	2021-09-19	2026-08-31	91.0	496	0
PANGKALANBUN2	Pangkalan Bun	2021-09-24	2026-08-31	91.2	1,353	0
MAROS	Maros	2021-10-13	2026-08-31	91.3	2,070	0
SAMARINDA2	Samarinda	2021-01-01	2026-08-31	91.8	1,181	0
JAMBI3	Kota Jambi	2021-01-01	2026-08-31	92.8	2,479	0
BENGKULU	Bengkulu	2021-08-19	2026-08-31	93.4	576	0
INDRAPURI2	Indrapuri	2021-10-04	2026-08-31	93.6	757	0
PEKANBARU2	Pekanbaru	2021-01-01	2026-08-31	93.6	2,547	0
JAMBI4	Muaro Jambi	2021-08-18	2026-08-31	94.3	1,742	0
KOTABARU	Kotabaru	2021-10-09	2026-08-31	94.7	52	0
MALANG	Malang	2021-09-23	2026-08-31	94.8	1,085	0
PALEMBANG4	Musi 2 Palembang	2021-08-21	2026-08-31	95.6	1,191	0
PESAWARAN	Pesawaran	2021-08-05	2026-08-31	95.8	890	0
SEMARANG	Semarang	2021-09-02	2026-08-31	95.8	1,609	0
PONTIANAK2	Mempawah	2021-09-09	2026-08-31	96.3	629	0
PALEMBANG3	Talang Betutu Palembang	2021-01-04	2026-08-31	96.4	928	0
KEMAYORAN3	Kemayoran	2021-08-25	2026-08-31	96.6	785	0
MEDAN2	Medan	2021-09-01	2026-08-31	97.1	946	0
MLATI	Sleman	2021-09-09	2026-08-31	97.4	286	0
SUPADIO	Kubu Raya	2021-09-09	2026-08-31	97.9	456	0
BATAM2	Batam	2021-09-14	2026-08-31	97.9	292	0
BANJARBARU2	Banjarbaru	2021-09-08	2026-08-31	98.2	234	0

A Technical methods and calculations

A.1 Notation, units, and forecast indexing

Let s index station, t the 00 UTC issue time, and $h \in \{3, 6, 12, 24, 48, 72\}$ h the direct forecast lead. Valid time is $v = t + h$, the quality-controlled observation is $y_{s,v}$ (µg m^{-3}), and the forecast is $\hat{y}_{s,t,h}$. Each lead has a separately fitted model. UTC is used for storage and splitting; local-clock features use each station's recorded UTC offset. CAMS conversion is

$$x^{\text{CAMS}}_{s,t,h}[\mu\text{g}\,\text{m}^{-3}] = 10^9 x^{\text{CAMS}}_{s,t,h}[\text{kg}\,\text{m}^{-3}]. \tag{1}$$

Observed $PM_{2.5}$ is valid for $0 \le y < 985$ µg m^{-3}; relative humidity for $0 < RH \le 100$%; and temperature for $-10 \le T \le 50$ °C. Invalid values become missing, never replacements. Monthly coverage is

$$C_{s,m} = 100\, n^{\text{valid}}_{s,m} / n^{\text{expected}}_{s,m}, \tag{2}$$

where expected hours lie inside the calendar-month/station-span intersection.

A.2 Predictor calculations

Historical lags are $L^{(k)}_{s,t} = y_{s,t-k}$ for $k = 0, 1, 2, 3, 6, 12, 24, 48, 72, 168$ h. Meteorological lags use $k = 0, 1, 3, 6, 12, 24$ h. For $W \in \{3, 6, 12, 24, 72, 168\}$ h and its finite set $A_{s,t,W}$,

$$\bar{y}_{s,t,W} = |A|^{-1} \sum_{u \in A} y_{s,u}, \qquad \sigma_{s,t,W} = \sqrt{|A|^{-1} \sum_{u \in A} (y_{s,u} - \bar{y})^2}, \tag{3}$$

$$M_{s,t,W} = \max_{u \in A} y_{s,u}, \qquad a_{s,t,W} = |A|/W. \tag{4}$$

At least one value is sufficient; no gap is filled. The age feature is time since the latest valid $PM_{2.5}$. The network mean excludes station s. For neighbours within 400 km, $w_{sj} = 1/\max(d_{sj}, 25)$ and

$$G_{s,t} = \frac{\sum_{j \ne s} w_{sj} y_{j,t}}{\sum_{j \ne s} w_{sj}}. \tag{5}$$

Distance is $d = 2R \arcsin \sqrt{\sin^2(\Delta\phi/2) + \cos\phi_s \cos\phi_j \sin^2(\Delta\lambda/2)}$, $R = 6371.0088$ km. Cycles use $\sin(2\pi q/P)$ and $\cos(2\pi q/P)$ for 24 h and 365.25 d. Coordinates, UTC offset, categories, and CAMS complete the predictors. Categories are one-hot encoded; numerical missingness is handled natively by the trees.

A.3 Baselines, learners, and selection

Persistence is $\hat{y}^{\text{pers}}_{s,t,h} = y_{s,t}$. Training climatology is the station–target-month–local-hour median, falling back to station–hour, network hour, then overall training median. The boosted prediction is

$$F_M(\mathbf{x}) = F_0(\mathbf{x}) + \eta \sum_{m=1}^{M} f_m(\mathbf{x}). \tag{6}$$

LightGBM minimizes $\sum_i |y_i - F_M(\mathbf{x}_i)|$ with learning rate 0.035, at most 1400 trees, 63 leaves, minimum 120 child cases, 0.85 row/feature subsampling, and L1/L2 penalties 0.1/0.5. XGBoost uses absolute error, learning rate 0.04, at most 1200 histogram trees, depth 8, minimum child weight 20.0, 0.85 subsampling, and L1/L2 penalties 0.1/1.0. Early stopping patience is 80 rounds. Predictions are clipped at zero.

Selection minimizes the mean of six station-balanced 2025 validation MAEs; CAMS LightGBM is preferred if within 1% of the minimum. The selected model is opened once on 2026. The expanding-window training diagnostic uses three later-than-fit blocks; +24 h station-balanced out-of-fold MAE is 10.46 µg m^{-3}. Later 2025 hyperparameter selection makes this diagnostic non-independent for selection.

A.4 Accuracy metrics and station balancing

With $e_i = \hat{y}_i - y_i$,

$$\text{MAE} = n^{-1}\sum_i |e_i|, \qquad \text{RMSE} = \sqrt{n^{-1}\sum_i e_i^2}, \qquad \text{Bias} = n^{-1}\sum_i e_i, \tag{7}$$

$$r = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2 \sum_i (\hat{y}_i - \bar{\hat{y}})^2}}, \qquad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{8}$$

Station-balanced metrics equal $S^{-1}\sum_s m_s$. Station skill is $100(1 - \text{MAE}_{model,s}/\text{MAE}_{pers,s})$ and lead skill is its station mean. At +24 h the model and persistence MAEs are 10.84 and 13.11 µg m^{-3}, while displayed skill is 17.6%; it is the mean of station-specific ratios, not the ratio of mean MAEs.

A.5 Prediction intervals

Quantile models minimize $\rho_\tau(u) = u[\tau - \mathbb{1}(u < 0)]$, $u = y - \hat{q}_\tau(\mathbf{x})$, for $\tau = 0.1, 0.5, 0.9$. Quantiles are fitted on 2023–2024, tree counts selected on January–June 2025, and raw quantiles sorted. July–December 2025 gives

$$E_i = \max(\hat{q}_{0.1,i} - y_i,\ y_i - \hat{q}_{0.9,i}). \tag{9}$$

For $1 - \alpha = 0.8$, $p = \min[\lceil (n_c + 1)(1 - \alpha)\rceil / n_c, 1]$, $c_h = \max[Q_p(E), 0]$, and $[L_i, U_i] = [\max(0, \hat{q}_{0.1,i} - c_h), \max(0, \hat{q}_{0.9,i} + c_h)]$. At +24 h, $n_c = 4,395$ and $c_h = 0.312$ µg m^{-3}. Coverage is $100n^{-1}\sum_i \mathbb{1}(L_i \leq y_i \leq U_i)$; width is $n^{-1}\sum_i (U_i - L_i)$; and

$$IS_i = (U_i - L_i) + \frac{2}{\alpha}(L_i - y_i)\mathbb{1}(y_i < L_i) + \frac{2}{\alpha}(y_i - U_i)\mathbb{1}(y_i > U_i). \tag{10}$$

For +24 h, 6,084 test intervals yield 78.0% coverage, 31.7 µg m^{-3} mean width, and 53.4 µg m^{-3} mean interval score.

A.6 Events, resampling, transfer, and diagnostics

Each station–lead event threshold is its training q90. POD $= H/(H + M)$, FAR $= F/(H + F)$, and CSI $= H/(H + M + F)$. At +24 h, $H = 357$, $M = 423$, and $F = 148$, giving 45.8%, 29.3%, and 38.5%.

Absolute-error improvements are averaged by station–ISO-week, then 1,000 cluster samples of equal size are drawn with replacement; 2.5th/97.5th percentiles form the interval. The all-lead CAMS ablation is 0.178 µg m^{-3} [0.119, 0.242] across 945 station-weeks. Five seeded station folds omit held-station identity. The shorter-window sensitivity uses 2024 only. Shift uses standardized mean difference and a 10-training-decile population-stability index. Residual is $\hat{y} - y$. LightGBM split count is normalized as $100 I_j / \sum_k I_k$ within lead and averaged across leads; it is non-causal.

A.7 Exact derivation of report figures

Display	Population	Calculation and display rule
Figure 1	Station-month hours	Valid/expected hourly coverage inside station span; grey means unavailable.
Figure 2	Common test cases	Station MAE first, then equal station mean; identical cases across models.
Figure 3	Station and lead	$100(1 - \text{MAE}_m/\text{MAE}_p)$; diverging scale centred at zero.

Display	Population	Calculation and display rule
Figure 4	+24 h daily values	Median across available stations; no smoothing; absent days remain gaps.
Figure 5	Paired test cases	Hex-bin count and 1:1 line; pooled 99.5th-percentile display limit only.
Figure 6	Valid intervals	Inclusion percentage and arithmetic mean width by lead.
Figure 7	Training-q90 events	POD, FAR, and CSI from test hits, misses, and false alarms.
Figure 8	Held stations	Five station folds; equal station aggregation; station identity omitted.
Figure 9	Selected trees	Normalized split count per lead, then six-lead mean; non-causal.
Figure 10	Test residuals	Mean forecast minus observation by UTC-derived target month and lead.
Appendix B	All stations	Unsmoothed +6, +24, +72 h sequences and calibrated q10–q90 band; no interpolation.

All summaries are computed before plotting from saved, checksummed evidence. No chart applies statistical smoothing; Figure 5 axis clipping never changes metrics.

B Integrated all-station independent-test time-series atlas

These nine plates are part of this report, not a separate document. They show all stations at +6, +24, and +72 h from 1 January–31 August 2026. Black is observed PM2.5, orange is the selected model, dashed grey is persistence, and blue shading is the calibrated nominal 80% interval. Lines are unsmoothed and missing values are not interpolated.

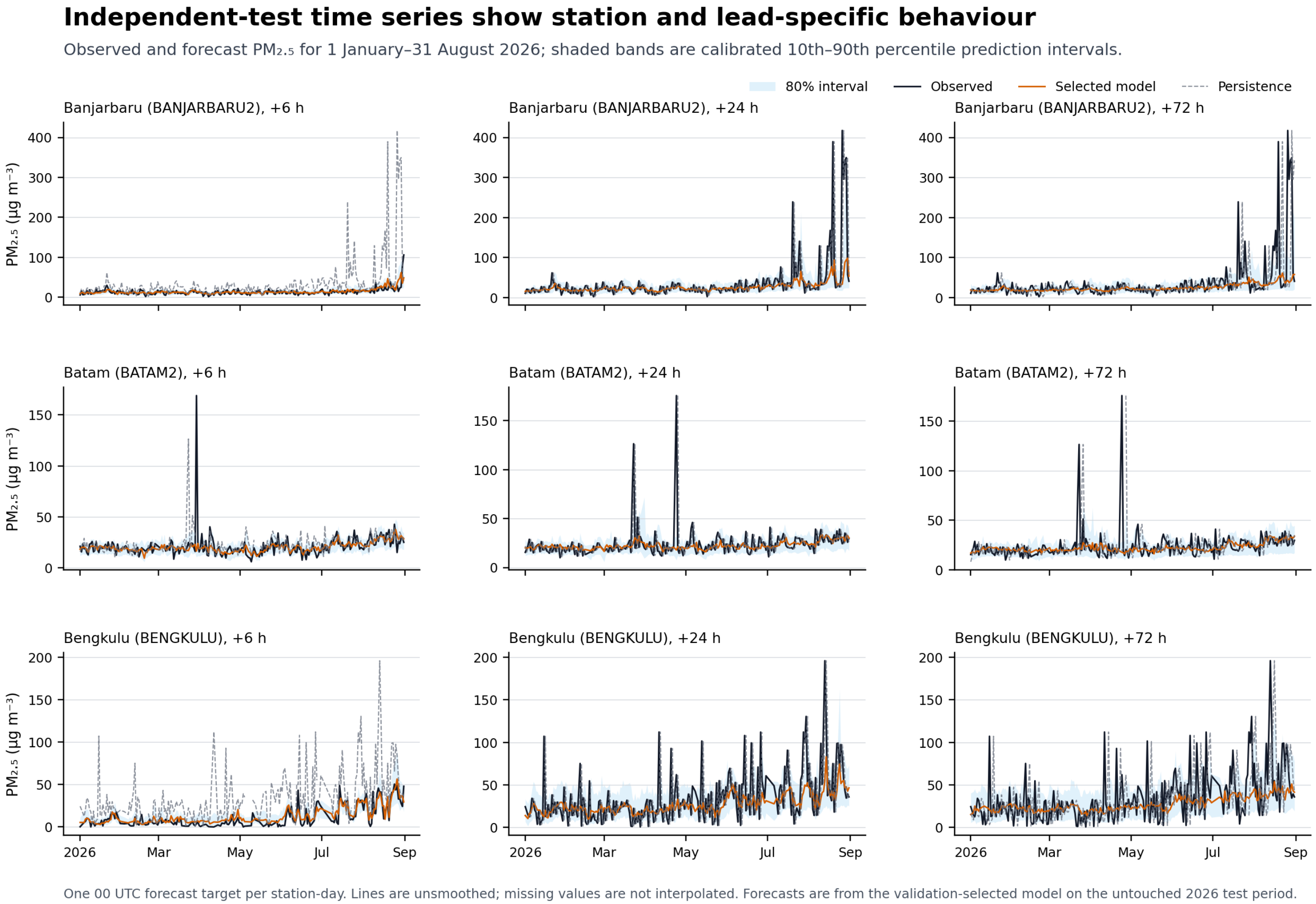

Figure 11: Atlas plate B1. Independent-test sequences for Banjarbaru (BANJARBARU2); Batam (BATAM2); Bengkulu (BENGKULU). Rows are stations; columns are +6, +24, and +72 h.

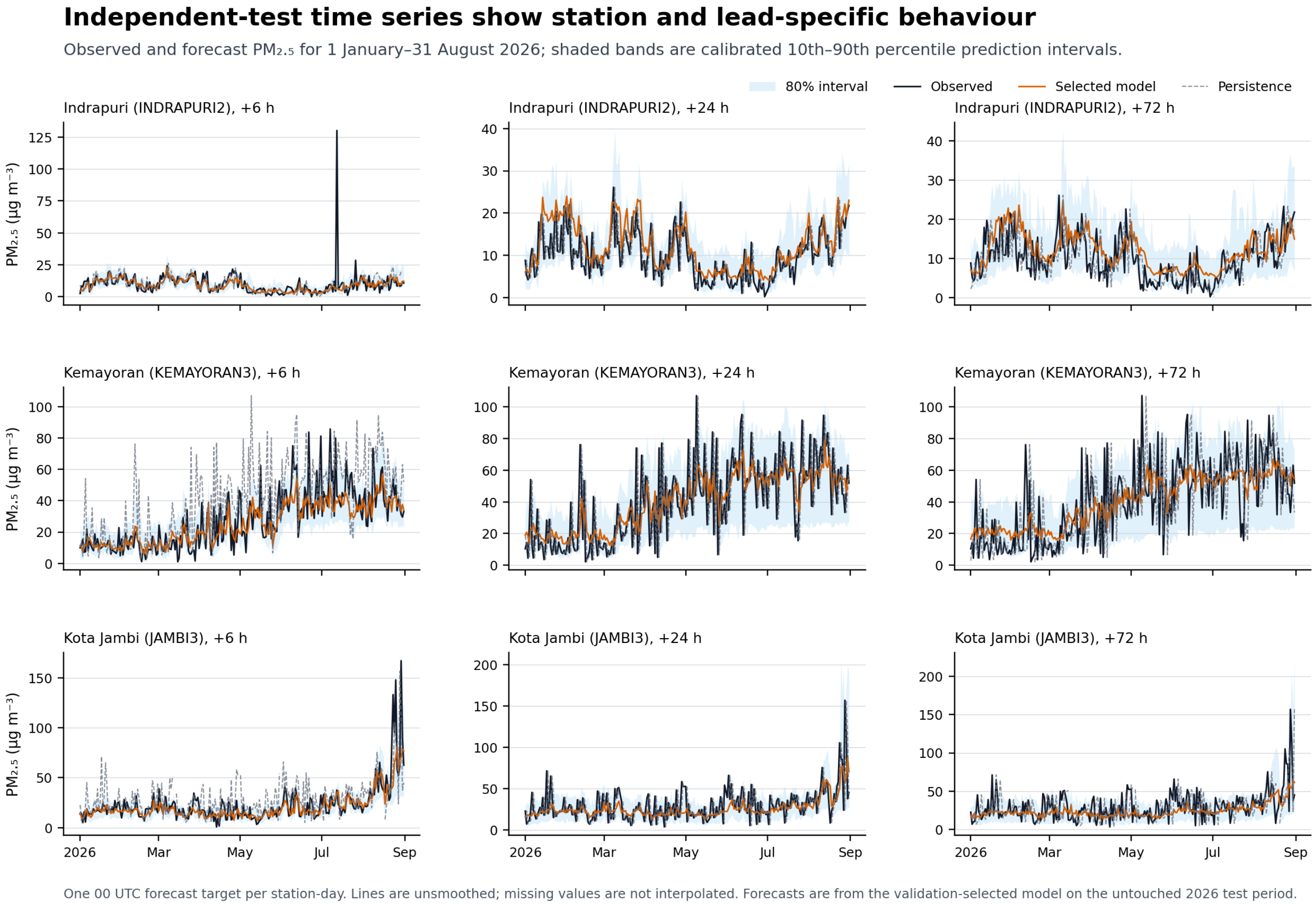

Figure 12: Atlas plate B2. Independent-test sequences for Indrapuri (INDRAPURI2); Kemayoran (KEMAYORAN3); Kota Jambi (JAMBI3). Rows are stations; columns are +6, +24, and +72 h.

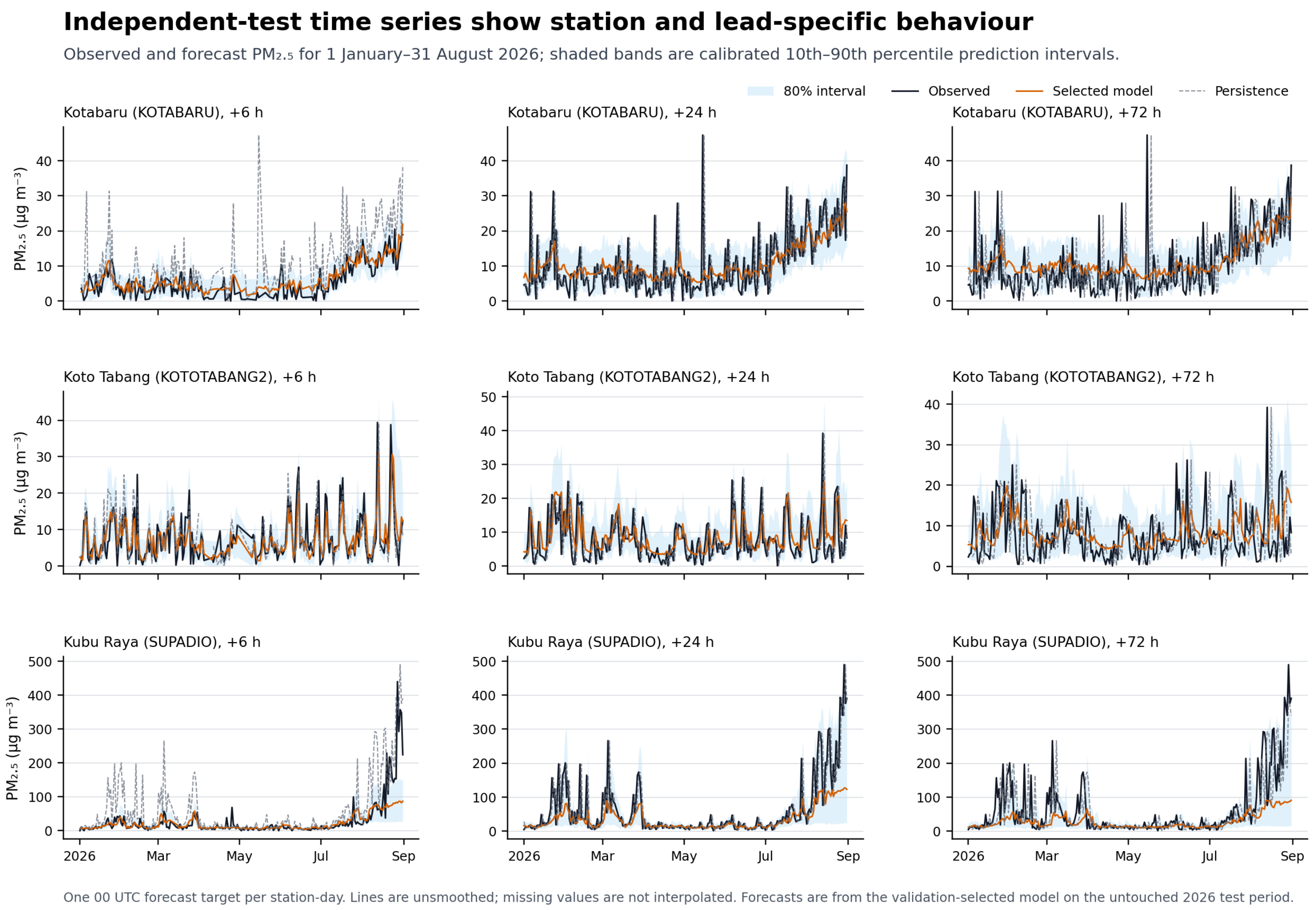

Figure 13: Atlas plate B3. Independent-test sequences for Kotabaru (KOTABARU); Koto Tabang (KOTOTABANG2); Kubu Raya (SUPADIO). Rows are stations; columns are +6, +24, and +72 h.

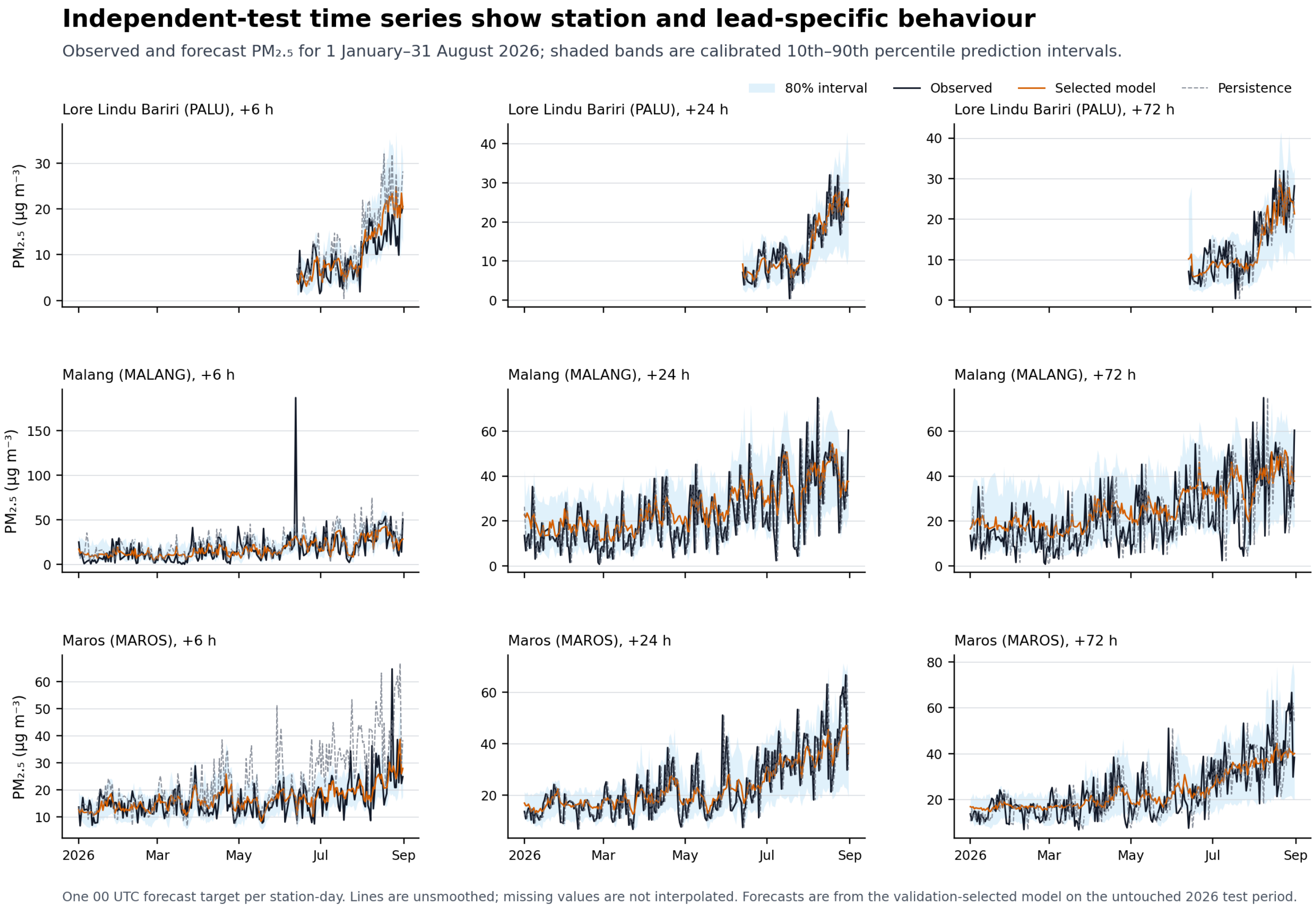

Figure 14: Atlas plate B4. Independent-test sequences for Lore Lindu Bariri (PALU); Malang (MALANG); Maros (MAROS). Rows are stations; columns are +6, +24, and +72 h.

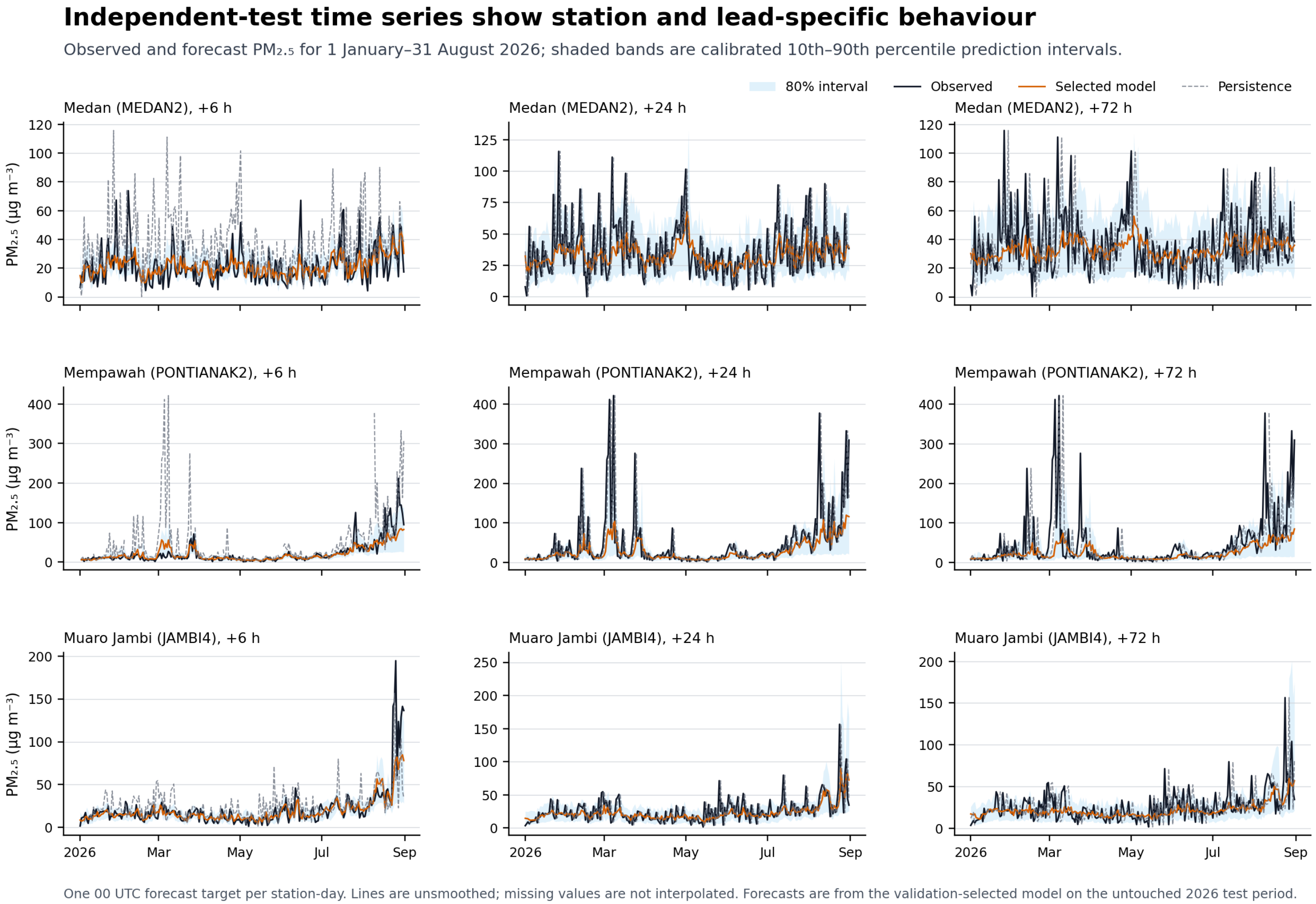

Figure 15: Atlas plate B5. Independent-test sequences for Medan (MEDAN2); Mempawah (PONTIANAK2); Muaro Jambi (JAMBI4). Rows are stations; columns are +6, +24, and +72 h.

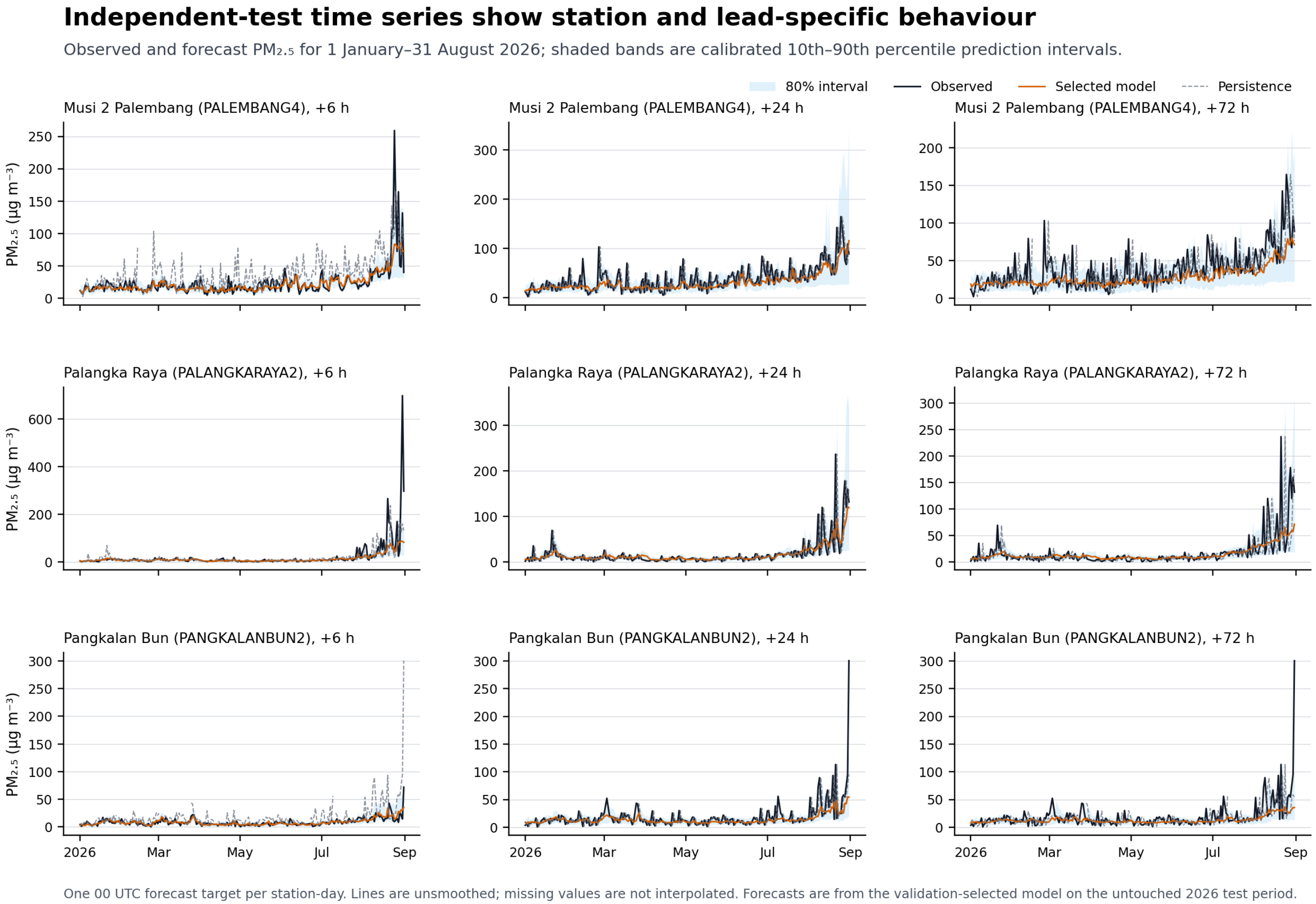

Figure 16: Atlas plate B6. Independent-test sequences for Musi 2 Palembang (PALEMBANG4); Palangka Raya (PALANGKARAYA2); Pangkalan Bun (PANGKALANBUN2). Rows are stations; columns are +6, +24, and +72 h.

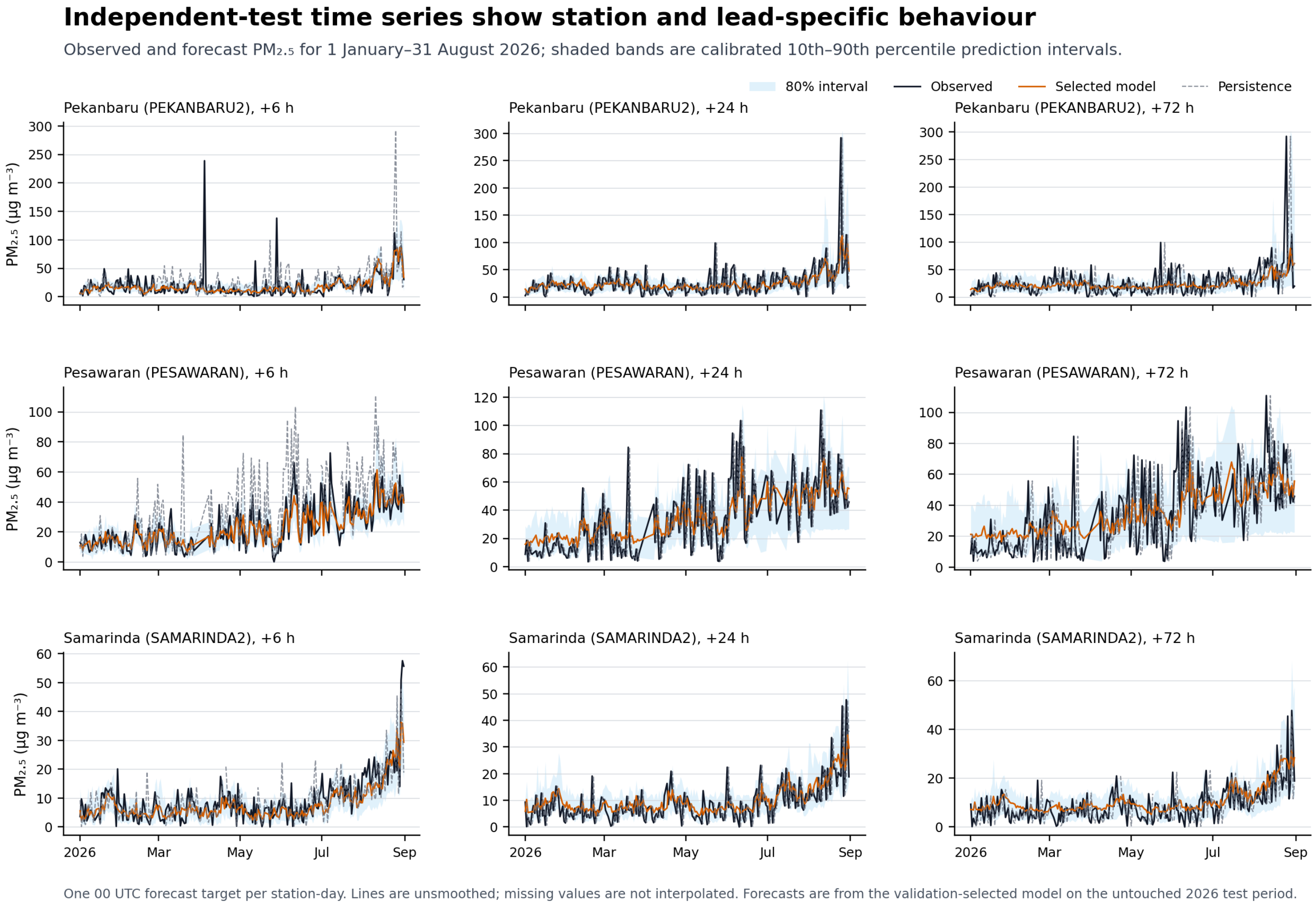

Figure 17: Atlas plate B7. Independent-test sequences for Pekanbaru (PEKANBARU2); Pesawaran (PESAWARAN); Samarinda (SAMARINDA2). Rows are stations; columns are +6, +24, and +72 h.

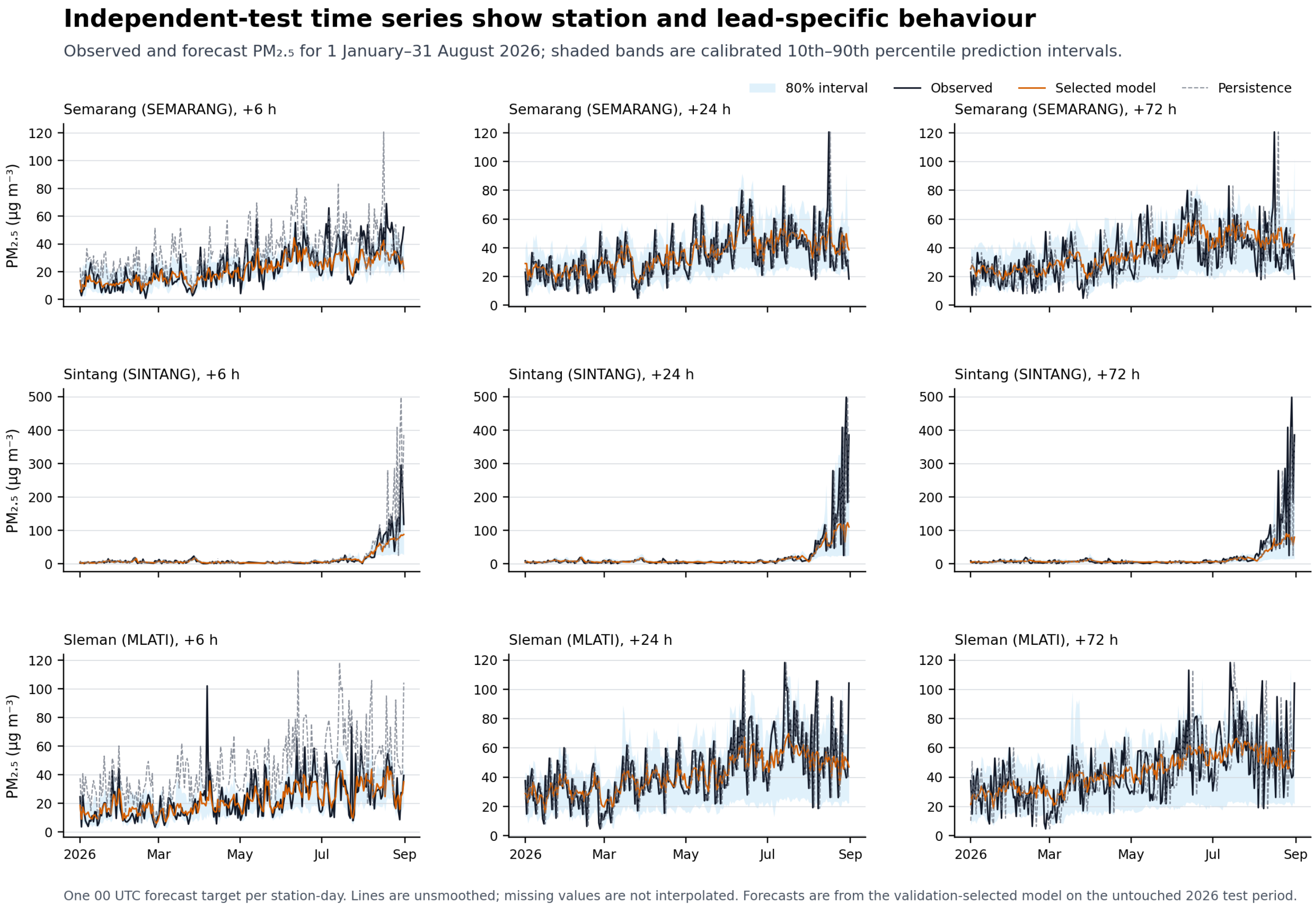

Figure 18: Atlas plate B8. Independent-test sequences for Semarang (SEMARANG); Sintang (SINTANG); Sleman (MLATI). Rows are stations; columns are +6, +24, and +72 h.

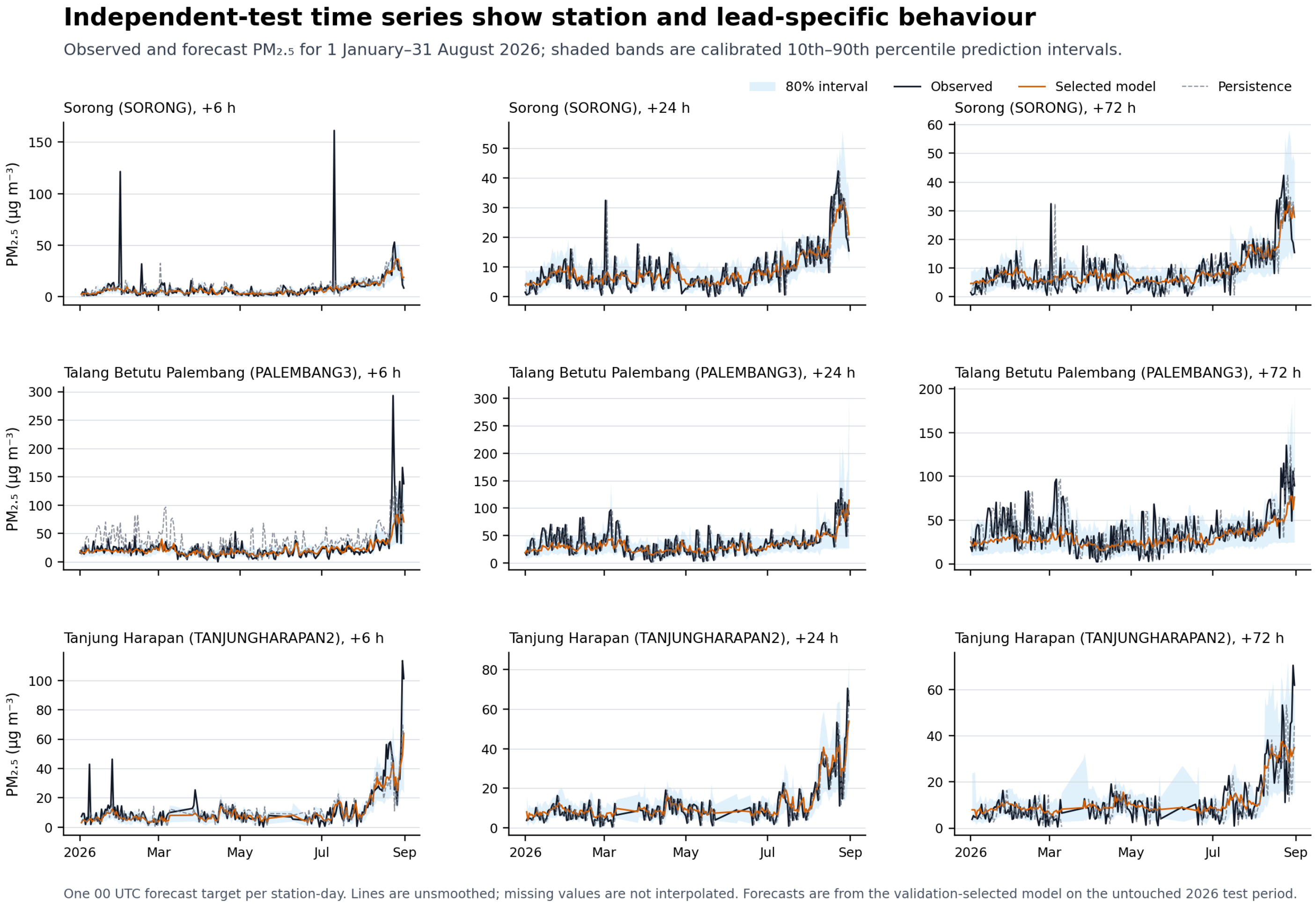

Figure 19: Atlas plate B9. Independent-test sequences for Sorong (SORONG); Talang Betutu Palembang (PALEMBANG3); Tanjung Harapan (TANJUNGHARAPAN2). Rows are stations; columns are +6, +24, and +72 h.

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